2026Year 2nd Semester Syllabus

Created Date	2026-07-22 12:38:53	Last-Modified	2026-07-22 13:00:58
Course Title	SYSTEM PROGRAMMING	Course Code-Section	CAS3107-01
Credit/Time/ Experiment,Lab,Practical Technique Time	3/Wed7,Fri7,8	Department	School of Computing
Time	Wed7,Fri7,8	Location	EngHD508
Exam Date & Time	Midterm exam	Final exam	
Class Language	English	Evaluation Type	Absolute evaluation

Instructor's Profile	Name	Park Seonghoon	Contact Information	Telephone	
	Department	DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING		Mail	seonghoon.park@yonsei.ac.kr
	Office			Interview information	

TA's Name & Contact Information	Name		Contact Information	Telephone	

Course Description Brief Introduction of the Course	This course covers the fundamental concepts and practical aspects of operating system kernel programming, with a particular focus on the Linux kernel. Key topics include the data structures, algorithms, and programming techniques used in kernel development.			
Course Goals	1.	Korean	운영체제 커널 프로그래밍의 기본 개념과 실무적 원리를 이해한다.	30%
		English	Understand the fundamental concepts and practical aspects of operating system kernel programming.	
	2.	Korean	리눅스 커널에서 사용되는 주요 자료구조, 알고리즘 및 프로그래밍 기법을 학습한다.	40%
		English	Explore key data structures, algorithms, and programming techniques used in the Linux kernel.	
	3.	Korean	리눅스 커널 프로그래밍에 대한 실무 경험을 쌓는다.	30%
		English	Gain hands-on experience in Linux kernel programming.	
	4.	Korean		0%
		English		
	5.	Korean		0%
		English		

Core Competencies	The total measurable competencies must be 100%. Each course objective should set the competency as 25%. The core and major competencies should equal at least 50%.		
	프로그래밍능력 50%	논리적사고 25%	문제해결능력 25%
Sub-Competencies/Learning Unit1			
Sub-Competencies/Learning Unit2			
Sub-Competencies/Learning Unit3			
Core Competencies(Liberal Arts)Major competency(	**Must reflect the interrelationship between core competencies (elective courses) and major competencies (major studies).**		
프로그래밍능력			
Sustainable Development Goals			

Average Recommended Amount of Learning per	Average Reading Volume		Average amount of writing(Based on A4)	

Course Methods (%) Total Amount 100	Lecture	Practice Training	Presentation	Dabate	Team Project
	80%	20%	0%	0%	0%
Course Methods 2 Select Relevant Items	PBL Subject	Capstone Design	CBL, Social Innovation Course	Flipped Classroom	Work Experience,Internsh

Grading Policy(%) Total Amount 100 Free Input for Other Information	Midterm exam	Final exam	Quiz	Individual Assignment	Team Assignment	Attendance	Others
	30%	30%	0%	30%	0%	10%	0%

Assignment/ Report, Project Guide	Title of Assignment/Project Name, and Method of Filling Out	Submission Deadline	Type of Submission and Method
Prerequisite	Operating Systems (운영체제)	Online Course Address	

Course Material	Course Material Name	Author	Publisher	Publish Year	ISBN

Main Learner Precautions	Students are expected to have completed an Operating Systems course and have a solid understanding of C programming. The use of generative AI tools is partially permitted for assignments.
Attatchment	

Weekly Plan

week	Period	Weekly Topic & Contents	Remarks
1	2026-09-01 2026-09-07	Introduction to Linux	(9.1.) Fall semester classes begin (9.3. - 9.7.) Course add and drop period
2	2026-09-08 2026-09-14	Linux process programming (1)	
3	2026-09-15 2026-09-21	Linux processing programming (2)	
4	2026-09-22 2026-09-28	Process scheudling in Linux	(9.24. - 9.25.) Chuseok Holidays 09.24 추석, 09.25 추석, 09.26 추석
5	2026-09-29 2026-10-05	Interrupt handling and processing (1)	(10.5.) Alternative Holiday of the National foundation Day 10.03 개천절, 10.05 대체공휴일(개천절)
6	2026-10-06 2026-10-12	Interrupt handling and processing (2)	(10.7.) First third of the semester ends (10.9.) Hangul Proclamation Day 10.09 한글날
7	2026-10-13 2026-10-19	Time processing in Kernel	(10.13. - 10.15.) Course withdrawal period
8	2026-10-20 2026-10-26	Mid-term examination	(10.20. - 10.26.) Midterm Examinations
9	2026-10-27 2026-11-02	System call and signal handling	(10.27. - 10.29.) Application Period for S/U evaluation
10	2026-11-03 2026-11-09	Kernel synchronization	
11	2026-11-10 2026-11-16	Linux memory management (1)	(11.13.) Second third of the semester ends
12	2026-11-17 2026-11-23	Linux memory management (2)	
13	2026-11-24 2026-11-30	Process address space management	
14	2026-12-01 2026-12-07	Understading the Linux file system (1)	
15	2026-12-08 2026-12-14	Understading the Linux file system (2)	(12.8. - 12.14.) Self-study
16	2026-12-15 2026-12-21	Final examimantion	(12.15. - 12.21.) Final Examinations

Attendance Requirement

• Students who are absent for more than one-third of the total class hours will receive an F or NP grade regardless of examination results.
• Courses without midterm and/or final examinations will hold regular classes during the examination periods.

Support for Students with Disabilities(SWDs)

• SWDs can request accommodations related to lectures, assignments, or tests by contacting the course professor at the beginning of semester. However, accommodations may vary depending on the essentiality of lecture and discretion of professors.

[Lecture]
- Visual Impairment: alternative, braille, enlarged reading materials, note-taker
- Physical Impairment: alternative reading materials, access to classroom, note-taker, assigned seat
- Hearing Impairment: note-taker/stenographer, recording lecture
- Intellectual Disability/Autism: note-taker

[Assignments and Test]
- Visual/Physical/Hearing Impairment: (reasonable) extra days for submission, alternative type of assignment, extended test time, alternative type of test, arranging separate test room, and proctors, test ghostwriter
- Intellectual Disability/Autism: (reasonable) extra days for submission, alternative type of assignment

Safety Precautions

• Students required to complete laboratory safety training must finish the online training before the semester begins and submit the certificate at the first class session. Failure to do so may result in restricted class participation.

• Students in physical activity courses should complete appropriate warm-up exercises before class. Those with cardiovascular or chronic respiratory conditions are advised to consult a physician before participating.

Course Administration

• Notices regarding class cancellations, make-up classes, classroom changes, and other course-related matters may be posted on LearnUs. For updates, please check LearnUs regularly.